

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics III

Topics in Analysis (3TN)

Some Examples of Banach Spaces

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0.1 The ℓ^p -spaces with $1 \leq p \leq \infty$

Let

$$\ell^\infty(\mathbb{C}) := \left\{ x = (x_n) \subset \mathbb{C} : \sup_{n \in \mathbb{N}} |x_n| < \infty \right\}, \quad \|x\|_\infty := \sup_{n \in \mathbb{N}} |x_n| \quad \forall x = (x_n) \in \ell^\infty(\mathbb{C})$$

and for $1 \leq p < \infty$,

$$\ell^p(\mathbb{C}) := \left\{ x = (x_n) \subset \mathbb{C} : \sum_{n=0}^{\infty} |x_n|^p < \infty \right\}, \quad \|x\|_p := (\sum_{n=0}^{\infty} |x_n|^p)^{1/p} \quad \forall x = (x_n) \in \ell^p(\mathbb{C}).$$

Theorem 0.1 $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are norms on $\ell^1(\mathbb{C})$ and $\ell^\infty(\mathbb{C})$ respectively and $(\ell^1(\mathbb{C}), \|\cdot\|_1)$ and $(\ell^\infty(\mathbb{C}), \|\cdot\|_\infty)$ are Banach spaces.

Proof: Show that $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are norms on $\ell^1(\mathbb{C})$ and $\ell^\infty(\mathbb{C})$ respectively!!!

Now, let (x^k) (with $x^k = (x_n^k)$) be a Cauchy sequence in $(\ell^1(\mathbb{C}), \|\cdot\|_1)$. That is,

$$\begin{aligned} \forall \varepsilon > 0 \quad \exists k_\varepsilon \in \mathbb{N} \text{ such that } \forall k, l \in \mathbb{N} \quad k, l \geq k_\varepsilon &\Rightarrow \|x^k - x^l\|_1 = \sum_{n=0}^{\infty} |x_n^k - x_n^l| < \varepsilon. \quad (*) \\ &\Rightarrow |x_n^k - x_n^l| < \varepsilon \quad \forall n \in \mathbb{N}. \end{aligned}$$

Hence, for every $n \in \mathbb{N}$, the sequence $(x_n^k)_{k \in \mathbb{N}}$ is Cauchy in $\mathbb{C}$ (complete) and hence, converges to some $x_n \in \mathbb{C}$.

Set $x = (x_n)$. Let us first show that $\sum_{n=0}^{\infty} |x_n| < \infty$, that is, $x \in \ell^1(\mathbb{C})$ and that (x^k) converges to x in $(\ell^1(\mathbb{C}), \|\cdot\|_1)$. It follows from $(*)$ that

$$\forall \varepsilon > 0 \quad \exists k_\varepsilon \in \mathbb{N} \text{ such that } \forall k, l \in \mathbb{N} \quad k, l \geq k_\varepsilon \Rightarrow \sum_{n=0}^m |x_n^k - x_n^l| < \varepsilon \quad \forall m \in \mathbb{N} \quad (**)$$

Now, passing to the limit in $(**)$ as $l \rightarrow \infty$, we get that

$$\forall \varepsilon > 0 \quad \exists k_\varepsilon \in \mathbb{N} \text{ such that } \forall k \in \mathbb{N} \quad k \geq k_\varepsilon \Rightarrow \sum_{n=0}^m |x_n^k - x_n| < \varepsilon \quad \forall m \in \mathbb{N} \quad (1)$$

and hence,

$$\sum_{n=0}^m |x_n| \leq \sum_{n=0}^m |x_n^{k_\varepsilon}| + \sum_{n=0}^m |x_n^{k_\varepsilon} - x_n| < \|x^\varepsilon\|_1 + \varepsilon \quad \forall m \in \mathbb{N}. \quad (2)$$

Now passing to the limit in (2) and in (1) as $m \rightarrow \infty$, we get that $x = (x_n) \in \ell^1(\mathbb{C})$ and that (x^k) converges to x in $\ell^1(\mathbb{C}), \|\cdot\|_1$ respectively.

Show that $(\ell^\infty(\mathbb{C}), \|\cdot\|_\infty)$ is a Banach space!!!! ■

Let $1 < p < \infty$. To show that $\|\cdot\|_p$ defines a norm on $\ell^p(\mathbb{C})$, we need the Hölder and Minkowski inequalities. The proof of the Hölder inequality uses another important inequality called the Young inequality that we will first state and prove.

Lemma 0.2 (The Young inequality)

Let $p > 1$ and $q > 1$ be real numbers satisfying $\frac{1}{p} + \frac{1}{q} = 1$. If $a > 0$ and $b > 0$, then

$$ab \leq \frac{1}{p}a^p + \frac{1}{q}b^q.$$

Proof: Consider the function $f(x) = x^{p-1}$ as defined in $[0, \infty)$ and take $x = a$ and $y = b$. Assume that $a^{p-1} < b$. We first find that the area I of the domain below the graph of f above the interval $[0, a]$ is equal to

$$I = \int_0^a x^{p-1} dx = \frac{1}{p}a^p.$$

We also find that the area II of the domain above the graph of f and below the line $y = b$ is given by

$$II = \int_0^b \int_0^{y^{\frac{1}{p-1}}} 1 dx dy = \int_0^{b^{\frac{1}{p-1}}} \int_{x^{p-1}}^b 1 dy dx = \frac{1}{q}b^q.$$

According to the figure below, the area ab of the rectangle with sides a and b is less than the sum of the areas (I and II), which establishes the Young inequality. The proof works in the same way for the case $b \leq a^{p-1}$.

Notice that the Young inequality can also be established through the convexity property of the function e^x . We will study the notion of convexity later. ■

We are now in a position to state and prove the Hölder inequality.

Proposition 0.3 (The Hölder inequality)

Let $1 \leq p, q \leq \infty$ with $\frac{1}{p} + \frac{1}{q} = 1$ (convention: $q = \infty$ when $p = 1$). Then for every $x = (x_n) \in \ell^p(\mathbb{C})$ and every $y = (y_n) \in \ell^q(\mathbb{C})$, we have that $xy = (x_n y_n) \in \ell^1(\mathbb{C})$ and

$$\|xy\|_1 \leq \|x\|_p \|y\|_q \Leftrightarrow \sum_{n=0}^{\infty} |x_n y_n| \leq \left(\sum_{n=0}^{\infty} |x_n|^p \right)^{1/p} \left(\sum_{n=0}^{\infty} |y_n|^q \right)^{1/q}.$$

Proof: The inequality is trivial if $p = 1$ (and hence $q = \infty$) or viceversa. In fact,

$$\|xy\|_1 = \sum_{n=0}^{\infty} |x_n y_n| \leq \sum_{n=0}^{\infty} \left[|x_n| \sup_{n \in \mathbb{N}} |y_n| \right] = \sup_{n \in \mathbb{N}} |y_n| \sum_{n=0}^{\infty} |x_n| = \|x\|_1 \|y\|_{\infty}.$$

Now assume that $1 < p, q < \infty$. Notice that the inequality is trivial if $x_n = 0$ or $y_n = 0$ for every $n \in \mathbb{N}$. So, we assume that $x_n \neq 0$ and $y_n \neq 0$ for every $n \in \mathbb{N}$ (which is not restrictive as the indices n such that $x_n = 0$ or $y_n = 0$ do not have any effect in the series).

Step 1: Assume that $\|x\|_p = \|y\|_q = 1$ and let us show that $\|xy\|_1 \leq 1 = \|x\|_p \|y\|_q$. To this aim, we will use the Young inequality in Lemma 0.2 as follows: we take for every $n \in \mathbb{N}$, $a = |x_n|$, $b = |y_n|$ and get

$$|x_n y_n| \leq \frac{1}{p} |x_n|^p + \frac{1}{q} |y_n|^q.$$

Hence,

$$\begin{aligned} \|xy\|_1 &= \sum_{n=0}^{\infty} |x_n y_n| \leq \frac{1}{p} \sum_{n=0}^{\infty} |x_n|^p + \frac{1}{q} \sum_{n=0}^{\infty} |y_n|^q \\ &= \frac{1}{p} \|x\|_p^p + \frac{1}{q} \|y\|_q^q = \frac{1}{p} + \frac{1}{q} = 1 = \|x\|_p \|y\|_q. \end{aligned}$$

Step 2: The general case: Let $u = (u_n)$ and $v = (v_n)$ with

$$u_n := \frac{x_n}{\|x\|_p} \quad \text{and} \quad v_n := \frac{y_n}{\|y\|_q} \quad \forall n \in \mathbb{N}.$$

Hence $\|u\|_p = \|v\|_q = 1$ and by Step 1, we get that

$$\|uv\|_1 \leq 1 \quad \Rightarrow \quad \sum_{n=0}^{\infty} \left| \frac{x_n}{\|x\|_p} \frac{y_n}{\|y\|_q} \right| \leq 1 \quad \Rightarrow \quad \frac{1}{\|x\|_p \|y\|_q} \sum_{n=0}^{\infty} |x_n y_n| \leq 1 \quad \Rightarrow \quad \|xy\|_1 \leq \|x\|_p \|y\|_q.$$

■

Now we are in a position to show that $\|\cdot\|_p$ defines a norm on $\ell^p(\mathbb{C})$ for $1 < p < \infty$ by establishing the triangle inequality for $\|\cdot\|_p$ called the Minkowsky inequality.

Proposition 0.4 (The Minkowsky inequality)

Let $x = (x_n)$ and $y = (y_n)$ be in $\ell^p(\mathbb{C})$. Then

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p. \quad (3)$$

Proof: The inequality is trivial if $\|x + y\|_p = 0$. So, we will assume that $\|x + y\|_p \neq 0$. Now,

$$\|x + y\|_p^p = \sum_{n=0}^{\infty} |x_n + y_n|^p = \sum_{n=0}^{\infty} |x_n + y_n|^{p-1} |x_n + y_n| \leq \sum_{n=0}^{\infty} |x_n + y_n|^{p-1} |x_n| + \sum_{n=0}^{\infty} |x_n + y_n|^{p-1} |y_n|. \quad (4)$$

Now, we are going to apply the Hölder inequality to each of the sums $\sum_{n=0}^{\infty} |x_n + y_n|^{p-1} |x_n|$ and $\sum_{n=0}^{\infty} |x_n + y_n|^{p-1} |y_n|$ (in fact, we only apply the Hölder inequality to the first sum and then swap x_n and y_n in order an application oh the Hölder inequality to the second sum). Therefore,

$$\begin{aligned}
 \sum_{n=0}^{\infty} |x_n + y_n|^{p-1} |x_n| &\leq \left(\sum_{n=0}^{\infty} |x_n|^p \right)^{1/p} \left(\sum_{n=0}^{\infty} [|x_n + y_n|^{p-1}]^q \right)^{1/q} \\
 &= \left(\sum_{n=0}^{\infty} |x_n|^p \right)^{1/p} \left(\sum_{n=0}^{\infty} [|x_n + y_n|^{p-1}]^{\frac{p}{p-1}} \right)^{\frac{p-1}{p}} \\
 &= \left(\sum_{n=0}^{\infty} |x_n|^p \right)^{1/p} \left(\sum_{n=0}^{\infty} |x_n + y_n|^p \right)^{\frac{p-1}{p}} = \|x\|_p \|x + y\|_p^{p-1}. \quad (5)
 \end{aligned}$$

Similary, by swapping x and y , we also get that

$$\sum_{n=0}^{\infty} |x_n + y_n|^{p-1} |y_n| \leq \|y\|_p \|x + y\|_p^{p-1}. \quad (6)$$

Therefore, we obtain from (4)-(6) that

$$\|x + y\|_p^p \leq \|x\|_p \|x + y\|_p^{p-1} + \|y\|_p \|x + y\|_p^{p-1} = (\|x\|_p + \|y\|_p) \|x + y\|_p^{p-1}.$$

Therefore,

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p$$

■

Theorem 0.5 Let $1 < p < \infty$. Then $\|\cdot\|_p$ is a norm on $\ell^p(\mathbb{C})$ and $(\ell^p(\mathbb{C}), \|\cdot\|_p)$ is a Banach space.

Exercise 0.6 Prove Theorem 0.5.

0.2 A charaterization of completeness on normed spaces

The following theorem gives a characterization of Banach spaces in terms of convergence of series.

Theorem 0.7 A normed space $(V, \|\cdot\|)$ is a Banach space if, and only if, it satisfies the following condition: for every sequence $\{v_h\} \subset V$ such that $\sum_{h=0}^{\infty} \|v_h\| < \infty$, the series $\sum_{h=0}^{\infty} v_h$ converges in V .

Proof: ($\Rightarrow$) Let $\{v_h\} \subset V$ be a sequence such that $\sum_{h=0}^{\infty} \|v_h\| < \infty$ and let $\{s_n\}$ be the sequence of partial sums. From the assumption and the fact

$$\|s_n - s_m\| = \sum_{h=m+1}^n \|v_h\| \leq \sum_{h=m+1}^{\infty} \|v_h\| \quad \forall n, m \in \mathbb{N} \text{ with } n > m,$$

it follows that $\{s_n\}$ is a Cauchy sequence, and hence since $(V, \|\cdot\|)$ is complete, $\{s_n\}$ does converges to a point in V . Therefore the series $\sum_{h=0}^{\infty} v_h$ converges in V .

($\Leftarrow$) Let $\{w_h\}$ be a Cauchy sequence in V . For any $k \in \mathbb{N}$, there exists $h_k \in \mathbb{N}$ (and it is not restrictive to assume $h_{k+1} > h_k$) such that $\|w_h - w_{h_k}\| < 2^{-k} \quad \forall h \geq h_k$. Now setting

$$\begin{cases} v_0 := w_{h_0} \\ v_{k+1} := w_{h_{k+1}} - w_{h_k}, \quad k \in \mathbb{N}; \end{cases}$$

it follows that $\{v_k\} \subset V$ and

$$\sum_{k=0}^{\infty} \|v_k\| = \|w_{h_0}\| + \sum_{k=0}^{\infty} \|w_{h_{k+1}} - w_{h_k}\| \leq \|w_{h_0}\| + \sum_{k=0}^{\infty} 2^{-k} < \infty.$$

Therefore, from assumption it follows that the series $\sum_{k=0}^{\infty} v_k$ converges in V , i.e., there exists $w \in V$ such that

$$w_{h_m} = \sum_{k=0}^m v_k \rightarrow w \quad \text{in } V.$$

Now since $\{w_h\}$ is itself a Cauchy sequence, it follows that all the sequence $\{w_h\}$ converges to w and hence $(V, \|\cdot\|)$ is a Banach space. ■

0.3 Compactness on finite dimensional normed spaces

Theorem 0.8 (*Compactness*) Let $(V, \|\cdot\|)$ be a normed space of finite dimension. Then a subset K of V is compact if and if K is closed and bounded.

Notice that Theorem 0.8 does not hold in infinite dimensional spaces. To show this, we will need the following lemma.

Lemma 0.9 (*F. Riesz*) Let $(V, \|\cdot\|)$ be a normed space over the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. Let Y and Z be two subspaces of V (of any dimension) and assume that Y is closed in V and that Y is a proper subspace of Z . Then for every $\theta \in (0, 1)$, there exists $z_\theta \in Z$ such that

$$\|z_\theta\| = 1 \quad \text{and} \quad \|z_\theta - y\| \geq \theta \quad \forall y \in Y.$$

Proof: Let $v \in Z \setminus Y$. Then, since Y is closed, it follows that

$$a := d(v, Y) = \inf_{y \in Y} \|v - y\| > 0. \quad (7)$$

From the definition of infimum, given $\theta \in (0, 1)$, there exists $y_\theta \in Y$ such that

$$a \leq \|v - y_\theta\| \leq \frac{a}{\theta}. \quad (8)$$

Set

$$z_\theta := c(v - y_\theta) \quad \text{with } c := \frac{1}{\|v - y_\theta\|}$$

Thus $\|z_\theta\| = 1$ and let us show that $\|z_\theta - y\| \geq \theta$ for every $y \in Y$. Notice that

$$\|z_\theta - y\| = \|c(v - y_\theta) - y\| = c\|v - y_\theta - c^{-1}y\| = c\|v - y_1\|$$

where $y_1 := y_\theta - c^{-1}y$. Notice that $y_1 \in Y$ and from the definition a , it follows $\|v - y_1\| \geq a$. So, using (7) and (8), we get

$$\|z_\theta - y\| = c\|v - y_1\| \geq ca = \frac{a}{\|v - y_\theta\|} \geq \frac{a}{a/\theta} = \theta.$$

and this completes the proof of Riesz's lemma. $\blacksquare$

Theorem 0.10 *If a normed space $(V, \|\cdot\|)$ has the property that the unit ball $B[0, 1]$ centered at 0_V is compact, then V has finite dimension.*

Proof: We prove this by contradiction. Assume that $B[0, 1]$ is compact and that $\dim(V) = \infty$. Let x_1 with $\|x_1\| = 1$ and let $X_1 = \langle x_1 \rangle$ which is a proper closed subspace of V . By Riesz's lemma, there exists $x_2 \in V$ with $\|x_2\| = 1$ such that

$$\|x_2 - x_1\| \geq \frac{1}{2}.$$

Let $X_2 := \langle x_1, x_2 \rangle$. Then X_2 is a proper closed subspace of V of dimension two. By Riesz's lemma, there exists $x_3 \in V$ with $\|x_3\| = 1$ and

$$\|x_3 - x\| \geq \frac{1}{2} \quad \forall x \in X_2$$

In particular,

$$\|x_3 - x_1\| \geq \frac{1}{2} \text{ and } \|x_3 - x_2\| \geq \frac{1}{2}.$$

Proceeding by induction, we get a sequence (x_n) of vectors in $B[0, 1]$ such

$$\|x_m - x_n\| \geq \frac{1}{2} \quad \forall n \neq m.$$

Thus, the sequence (x_n) does not have any converging subsequence and this contradicts the compactness of $B[0, 1]$. $\blacksquare$

In the next lecture, we start the study of linear continuous mappings between normed spaces.